

Second part: Open book

Total score: 10 pts.

You are warmly invited to be tidy and concise. Messy papers will be penalized.

6. In a transportation network there are a set of suppliers S , a set of customers C and a set of transshipment centers T . Let the set of nodes N of the network is given by $S \cup C \cup T$. The network is also defined by a set of directed connections between pairs of nodes $A \subset N \times N$. Each directed arc $(i, j) \in A$ represents the fact that a direct shipment can take place from i to j . In the network the same type of good must be transported from the suppliers to the customers, possibly passing through transshipment centers. For each node i the quantity s_i of good to be sent (if i is a supplier) or received r_i (if i is a customer) are known. We assume that the overall sum of the supplies is equal to the overall sum of the requests (i.e., $\sum_{i \in S} s_i = \sum_{i \in C} r_i$). Each connection $(i, j) \in A$ is characterized by a maximum amount u_{ij} that can be transported from i to j . The cost for transporting goods from i to j is c_{ij} and it is independent from the actual quantity transported. In practice there is a truck with capacity u_{ij} going from i to j and the cost of sending the truck is c_{ij} even if it is almost empty.

Formulate with a linear model the problem of planning the transportation of goods in such a way that the customer and suppliers requests are satisfied, considering the possible connections and their capacities. The objective is to minimize the overall transportation cost.

Variant: Let us assume that the transshipment centers have a cost for processing the transiting goods. The cost of processing one unit of good in center $i \in T$ is p_i . How does the formulation change?

7. Write the dual and the complementary slackness equations of the following problem having variables d, x_{ij}, y_i :

$$\min \quad d \quad (1)$$

$$d \geq \sum_{i=1}^n c_{ij} x_{ij} \quad j = 1, \dots, m \quad (2)$$

$$\sum_{j=1}^m a_{ij} x_{ij} \leq b_i y_i \quad i = 1, \dots, n \quad (3)$$

$$\sum_{i=1}^n y_i = 1 \quad (4)$$

$$d \geq 0 \quad (5)$$

$$y_i \geq 0 \quad i = 1, \dots, n \quad (6)$$

$$x_{ij} \text{ free in sign} \quad i = 1, \dots, n, j = 1, \dots, m \quad (7)$$

In the dual, use variables α_j associated with constraints (2), β_i associated with constraints (3), and γ associated with constraint (4)